Mehdi DJERBOUA



AI Engineer · M.Sc. in Artificial Intelligence

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Profile

AI engineer with an M.Sc. in Artificial Intelligence for Connected Industries (CNAM Paris, 2026), quadrilingual. During a 6-month research internship I built a full-stack digital twin of smart buildings (EV chargers, solar panels, batteries, building energy), with a natural-language assistant powered by a local LLM. Hands-on experience with large language models (local deployment, NL-to-SQL, multi-agent systems) and Python automation. Freelance developer since 2022.

Education

M.Sc. Computer Science – AI for Connected Industries (AI4CI) · CNAM Paris 2024 – 2026
Graduated 2026. Programme taught entirely in English.

B.Sc. Computer Science – Computer Systems · University of Blida 1 (USDB), Algeria 2020 – 2023
Key courses: artificial intelligence, algorithms, databases, networks and security, software engineering. Projects: lab management, Markov chain visualiser, pharmacy manager.

Baccalaureate in Mathematics · Algeria 2020

Experience

Research Intern · LVMT, joint lab of École des Ponts & Université Gustave Eiffel Mar – Aug 2026

- Led the design and development of **DEMI** end to end, turning an initial FIWARE prototype into a complete digital-twin platform for smart campus buildings (EV charging, solar panels, batteries, building energy).
- Engineered the full data layer: ingestion pipelines (FIWARE Orion-LD → QuantumLeap → CrateDB) for years of real charging and energy data, and an extended NGSI-LD data model covering solar, storage and building systems.
- Built the interactive web dashboard from scratch (campus map, energy flows, per-port charging timelines) with historical and live digital-twin modes, plus a validation tool that traces every update end to end.
- Integrated local AI models into the platform: a natural-language assistant that turns plain-English questions into database queries, and a live control and monitoring module with AI-generated explanations of system events; designed the benchmark used to evaluate and improve the assistant.
- Delivered a reproducible deployment: containerised the full stack with Docker and automated installation, start-up and data pipelines with scripts.

Freelance Developer · Self-employed 2022 – present

- Built and launched 5+ client websites, from online shops to service sites for small businesses, handling design, development, hosting and deployment on my own.
- Developed an automated email-campaign system for advertising, engineered to land reliably in the inbox instead of spam.

Summer Intern – Linux USB Gadgets · CNAM Paris 2025
Developed USB gadget drivers (HID, mass storage) in C and Python on embedded Linux; network configuration and system optimisation.

Translator & Interpreter · MAR-MOT, Polish piston-ring manufacturer 2023
Translated and interpreted for the company (Arabic, French, Polish, English), on the recommendation of the Embassy of Poland.

Election Assistant · Embassy of Poland 2024
Helped organise voting events at the embassy.

Selected Projects

Code Generation Crew – multi-agent LLM system · CNAM Paris, team of 3 2025

- Built a team of AI agents running on local models that write, review and test code together in a closed feedback loop, each round correcting the previous one until the program passes its tests.
- Showed that small local models working together produce reliable, working programs, including on complex tasks.

More projects (email automation, data visualisation, desktop and web apps): mehdiamine16.github.io

Technical Skills

AI & LLMs	Python, local LLM deployment (Ollama), prompt engineering, multi-agent systems (LangGraph), NL-to-SQL, machine learning, statistics, probability, linear programming, MATLAB
Data & infrastructure	Docker, FIWARE/NGSI-LD, CrateDB, MongoDB, FastAPI, SQL, Git/GitHub
Systems & programming	C, C++, Java, Bash, embedded Linux, networking, USB gadget drivers, JS, HTML/CSS

Languages & Interests

Languages: Arabic, French, Polish (native); English (professional – Master's taught, presented and written in English).

Interests: former Rubik's Cube champion (17 s average), judo (black belt), photography, chess.